G-Computation: Intuition and Math

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Intuitive Understanding

At its heart, **g-computation** (also known as the g-formula, standardization, or the substitution estimator) is a method to simulate a randomized controlled trial (RCT) using observational data. It answers the question: *"What would the average outcome in our entire population have been if everyone had received treatment A, versus if everyone had received treatment B?"*

Imagine we want to know the causal effect of a new fitness app (A) on weight loss (Y). We have data from a group of people, some of whom used the app $(A = 1)$ and some who didn't $(A = 0)$. We also know their age (L), which is a potential **confounder**—younger people might be both more likely to use a fitness app and more likely to lose weight, regardless of the app.

A simple comparison of weight loss between app users and non-users would be misleading because of this confounding by age. G-computation fixes this by creating a "standardized" population.

Here's the step-by-step logic:

1. **Model the World:** First, we use our observational data to build a statistical model that describes how the outcome (Y, weight loss) depends on the treatment (A, app use) and the confounder(s) (L, age). This model learns the `Weight Loss ~ App Use + Age` relationship from the data we have. A simple model could be a linear regression: $E[Y|A, L] = \beta_0 + \beta_1 A + \beta_2 L$.
2. **Create Hypothetical Worlds (Counterfactuals):** We take our *entire* study population and create two hypothetical copies:
   - **World 1 (Everyone gets treated):** In this copy, we change the `App Use` variable to 1 for *every single person*, but we keep their actual, observed age.
   - **World 2 (No one gets treated):** In this second copy, we change the `App Use` variable to 0 for *everyone*, again keeping their actual age.
3. **Predict and Average:** We now use the model from Step 1 to predict the outcome for everyone in these two new hypothetical worlds.

- For World 1, we predict each person's potential weight loss using their actual age but with `App Use` set to 1. We then average all these predictions. This gives us $E[Y^{a=1}]$, the estimated average weight loss if *everyone* in the population had used the app.
- For World 2, we do the same but with `App Use` set to 0. The average of these predictions gives us $E[Y^{a=0}]$, the estimated average weight loss if *no one* had used the app.

4. **Estimate the Causal Effect:** The Average Causal Effect (ATE) is simply the difference between the averages from our two hypothetical worlds:

$$\text{ATE} = \mathbb{E}\left[Y^{a=1}\right] - \mathbb{E}\left[Y^{a=0}\right]$$

This procedure gives a valid causal estimate because we have "broken" the link between the confounder (age) and treatment assignment. By forcing everyone to be treated (and then untreated), we have simulated an intervention and estimated what the outcome would be in a world where age no longer influences who gets the treatment.

Parametric vs. Non-Parametric G-Computation

- **Parametric G-Computation:** This is what was described above. It relies on a *parametric* statistical model (e.g., linear regression, logistic regression) to perform Step 1.
  - **Pro:** It's efficient and works well even with many continuous confounders.
  - **Con:** It's **model-dependent**. If your model of the world (`Y ~ A + L`) is wrong (e.g., the true relationship is non-linear but you used a linear model), your causal estimate will be biased.
- **Non-Parametric G-Computation:** This approach avoids assuming a specific model shape. Instead of fitting one big regression, it works by stratifying.
  1. Divide the population into strata based on the confounder(s) (e.g., age groups: 20-29, 30-39, etc.).
  2. Within each stratum, calculate the average outcome for the treated and the untreated.
  3. To get the overall $\mathbb{E}\left[Y^{a=1}\right]$, calculate a weighted average of the stratum-specific means for the treated, where the weights are the proportion of the *total population* in each stratum. Do the same for the untreated to get $\mathbb{E}\left[Y^{a=0}\right]$.

  - **Pro:** It's more flexible and less prone to model misspecification bias.
  - **Con:** It becomes impractical with many or continuous confounders (the "curse of dimensionality"), as strata become too small or empty.